

TRAVERS AGCM, AB, Canada

WMO: 718480

Lat: 50.3038N

Lon: 112.8626W

Elev: 3133

StdP: 13.11

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.0	-12.4	-23.7	1.7	-17.5	-18.2	2.3	-12.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.7	86.5	59.8	82.9	58.9	79.7	58.6	64.3	76.6	62.5	75.9	60.8	75.0	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.7	86.0	69.7	57.7	79.8	67.9	55.6	73.9	65.3	31.2	76.1	29.7	75.8	28.4	74.9	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-24.7	92.8	6.5	4.7	-29.4	96.1	-33.2	98.9	-36.8	101.5	-41.5	104.9		
(5)			-25.0	68.2	6.4	2.2	-29.6	69.8	-33.4	71.1	-37.0	72.3	-41.6	73.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.4	21.2	18.6	30.1	40.5	51.5	59.1	64.4	63.7	55.0	42.5	29.9	18.3
	DBStd	20.01	14.75	15.58	13.64	9.37	8.12	5.97	4.94	6.20	8.99	9.80	13.28	14.93
	HDD50	4684	891	879	619	302	80	4	0	2	50	267	608	982
	HDD65	8771	1356	1299	1083	735	423	192	71	98	317	697	1053	1447
	CDD50	1534	0	0	1	18	126	278	446	426	199	35	5	0
	CDD65	146	0	0	0	0	4	16	52	58	16	0	0	0
	CDH74	2388	0	0	0	15	127	262	812	887	280	5	0	0
	CDH80	673	0	0	0	1	22	60	224	288	77	1	0	0
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	PrecAvg	13.8	0.5	0.4	0.7	1.2	2.1	2.8	1.7	1.5	1.1	0.7	0.6	0.5
(15) Precipitation	PrecMax	21.0	1.0	0.9	2.2	2.7	3.6	7.9	4.0	3.7	4.3	1.9	1.5	1.1
	PrecMin	6.3	0.0	0.0	0.0	0.3	0.5	0.9	0.3	0.0	0.1	0.1	0.0	0.0
	PrecStd	2.6	0.2	0.2	0.4	0.6	0.9	1.5	1.0	0.9	0.9	0.4	0.4	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.5	53.1	62.1	76.1	83.0	87.0	90.6	91.8	86.6	73.2	64.7	50.8
		MCWB	41.7	41.9	44.9	51.6	55.3	62.2	61.1	60.6	57.8	52.5	48.0	39.2
	2%	DB	45.6	46.5	57.1	67.6	77.6	80.6	85.7	86.7	81.8	67.3	56.4	45.2
		MCWB	37.6	37.6	42.8	47.4	53.8	58.2	61.1	59.5	56.3	49.8	43.9	37.3
	5%	DB	42.4	41.8	52.6	61.9	72.2	76.4	81.7	82.9	76.6	62.7	51.2	41.3
		MCWB	35.9	34.7	40.6	44.8	51.7	57.5	60.6	59.3	54.7	47.6	41.6	34.6
	10%	DB	39.6	38.3	47.7	56.7	68.1	72.9	78.6	79.0	71.2	58.0	46.6	38.3
		MCWB	34.3	32.9	38.0	42.1	50.1	55.4	60.3	58.8	52.9	45.2	38.3	33.2
	0.4%	WB	41.4	42.8	46.1	51.7	56.5	64.8	67.6	65.1	60.3	54.2	48.6	41.8
		MCDB	49.5	52.4	58.1	72.5	77.2	78.7	77.5	79.6	78.7	69.0	63.0	48.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.2	38.1	43.8	48.4	54.8	61.6	65.3	62.8	58.0	51.2	45.1	37.8
		MCDB	44.7	45.8	55.5	64.7	74.1	75.2	77.3	78.8	76.2	64.3	55.3	44.2
	5%	WB	36.4	35.2	41.1	45.5	53.0	59.2	63.6	61.4	55.7	48.4	41.9	35.2
		MCDB	41.9	41.2	52.0	59.9	69.5	72.8	75.3	76.9	73.4	61.0	50.4	40.5
	10%	WB	34.5	32.9	38.5	43.0	51.3	57.1	62.0	59.9	53.9	46.1	38.9	32.9
		MCDB	39.6	38.1	47.0	55.6	64.9	68.9	74.3	74.7	69.4	57.1	46.0	38.2
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	18.2	19.4	22.0	22.2	21.7	24.7	25.3	23.3	21.4	18.1	16.4
		MCDBR	17.6	22.0	25.6	31.9	29.7	27.8	31.1	32.7	31.8	27.5	23.3	18.4
		MCWBR	11.9	14.5	14.2	15.3	12.4	11.5	12.1	11.4	12.2	13.5	13.6	12.6
	5% WB	MCDBR	16.7	21.1	24.6	30.3	26.3	25.1	25.9	28.0	29.6	26.5	23.2	19.4
		MCWBR	12.0	14.7	14.3	15.2	11.4	11.7	12.2	11.3	12.1	13.6	14.0	14.3
(40) Clear-Sky Solar Irradiance	taub		0.252	0.260	0.282	0.322	0.338	0.334	0.336	0.351	0.312	0.279	0.254	0.234
	taud		2.444	2.487	2.483	2.378	2.384	2.443	2.452	2.389	2.476	2.541	2.501	2.446
	Ebn at Noon		259	287	297	293	291	292	290	279	282	275	257	251
	Edh at Noon		18	23	28	35	36	34	33	34	28	22	17	16
(44) All-Sky Solar Radiation	RadAvg		388	701	1115	1534	1829	2010	2193	1750	1240	766	420	293
	RadStd		30	64	84	106	135	132	109	112	112	84	38	26

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (78 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page